

Farriha Afnan

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Education

Bangladesh University of Engineering and Technology (BUET)

Feb 2022 - June 2026

Bachelor of Science in Computer Science and Engineering

CGPA - 3.97/4.00 (Class rank - 7/120) • University Merit Scholarship • Dean's List Award

Research Interests

• Machine Learning • Deep Generative Models • Single-Cell Transcriptomics • Computational Biology • Graph Neural Networks • Health Informatics

Research Experience

Patient-Specific Clinical Deterioration Pathway Discovery from Longitudinal ICU Data

Research Assistant, UGC-ICSETEP Research and Development Grant (RDG)

Sep 2026 – Present

Supervisor: Dr. Atif Hasan Rahman, Professor, CSE, BUET; Dr. Sadia Sharmin, Associate Professor, CSE, BUET

Keywords: Graph Neural Networks • Temporal Graph Networks • Large Language Models • Electronic Health Records • GNN-LLM integration • Mortality Prediction

- Motivated by the fact that ICU mortality models output a risk score but not the **clinical path that leads to it**, the goal is to recover, for each patient, the sequence of interacting events that drives deterioration.
- Developing **multimodal, continuous-time event graphs** from irregular longitudinal ICU records, where clinical concepts (vitals, labs, interventions) are nodes and time-stamped clinical relationships are edges.
- Building a **temporal graph network** for mortality prediction with an edge-importance module that extracts **patient-specific deterioration pathways**.

scCrossGEN: A Generative Framework for Cross-Organ Single-Cell Transcriptomic Translation

Undergraduate Thesis

2025 – Present

Supervisor: Dr. Atif Hasan Rahman, Professor, CSE, BUET, Dhaka, Bangladesh

Collaborator: Dr. Md. Abul Hassan Samee, Associate Professor, Integrative Physiology, Baylor College of Medicine, Houston, Texas, USA

Keywords: Single-Cell Transcriptomics • Deep Generative Models • Variational Autoencoders • Adversarial Learning • Scanpy • DESeq2

- Developed **scCrossGEN**, a CycleGAN-inspired VAE-based **generative framework** with adversarial training that translates single-cell transcriptomes **between organs from cell-level unpaired data**.
- Generated a donor's cell populations in a hard-to-sample organ from their cells in an accessible one, outperforming **scGen** and **scPRAM** across multiple organ systems. Preprint is in preparation.

Multi-Omics Disease Prediction via Graph-Based Cross-Modal Imputation

Feb 2026

Supervisor: Nafis Tahmid, Lecturer, CSE, BUET, Dhaka, Bangladesh

- Developed a deep learning framework that jointly **predicts disease outcomes** and **reconstructs missing omics modalities** from single-omic inputs.
- Combined a generator network for cross-modal imputation with a **SynOmics-inspired graph neural network** that uses feature-level and bipartite relationships across omics.

Professional Experience

Bangladesh University of Engineering and Technology

Adjunct Lecturer(June 2026 - Present)

- Teaching 5 different courses, including Database, Computer Architecture, Computer Programming, Microcontroller, and Signals and Linear Systems in 7 sections, each with 30 students. • Supervised 29 projects.

Bangladesh University of Engineering and Technology

Research Assistant(September 2026 - Present)

- Patient-Specific Clinical Deterioration Pathway Discovery from Longitudinal ICU Data

Presidency University

Lecturer(June 2026 - August 2026)

- Taught 3 different courses, including OOP, Microcontroller and C Programming.

Bangladeshi Women in Computer Science and Engineering

Vice President(March 2025 - Present)

Event Director (August 2024 - March 2025)

Awards and Achievements

Dean's List Scholarship (2022, 2023, 2025)

Awarded for excellent performance in Level 1, Level 2 and Level 3 in BUET

University Merit Scholarship (2022 - 2026)

Acknowledged for outstanding results in different terms

BUET CSE Fest Hackathon (October 2024)

Best UI/UX prize (DevOps)

SKILL INNOVATION FEST, IUT (April 2024)

1st runner up - Project Showcasing Competition

BUET CSE Fest Hackathon (July 2023)

Among the top 15 qualifying teams (API and Cloud Service)

Government Talentpool & General Scholarship (2018 - 2025)

Recognized for outstanding performance in Higher Secondary Certificate and Secondary School Certificate Examination

Projects

Wondernest | June 2025 (team)

- An AI based learning website for kids, LLMs are integrated(OpenAi and Gemini)
- **Tools:** PostgreSQL, Spring Boot, React, GitHub Actions, Github, Docker, AWS for deployment.

Railway E-Ticketing System | October 2024 (team)

- An e-ticketing platform of Bangladesh Railway for BUET CSE fest 2024 Hackathon (Best UI/UX winner).
- **Tools:** MongoDB, Node.js, Next.js, HTML & CSS, Kubernetes, Github, Docker.

Deep Vision Voyager | January 2024 - February 2024 (team)

- Live streaming bot with object detection, wireless control through client-server network
- **Hardwares:** Raspberry Pi 4, motor, motor driver, webcam, Wi-fi adapter, servo motor & powerbank
- **Technologies:** YOLOv3 (Deep Learning Architecture for Object Detection), Python & Socket Programming

4-bit MIPS | February 2024 (team)

- 4 bit MIPS (Microprocessor without Interlocked Pipeline Stages) architecture using basic gates & AtMega32

4-bit ALU | December 2023 (team)

- 4 bit ALU (Arithmetic Logic Unit) using basic gates

Compiler for C | December 2023 - February 2024

- Compiler for a subset of C language
- **Tools:** Flex & Bison

Learnly | July 2023 - September 2023 (team)

- A web based tuition platform for connecting students and teachers, developed a user-friendly interface
- **Tools:** Oracle SQL, Next.js, HTML & CSS

Movie Database System | August 2022 - September 2022

- **Tools:** Java, JavaFX & networking

Crowd City | March 2022 - April 2022 (team)

- **Tools:** C & iGraphics

Coursework

Machine Learning, Operating System, AI, Information System Design, Concrete Mathematics, Computer Networks, Computer Architecture, Software Engineering, Compiler, Data Communication, Microcontroller, Theory of Computation, Database, Digital Logic Design, Numerical Methods, Discrete Mathematics, Data Structure and Algorithm, Bioinformatics, Simulation and Modeling, Fault Toleration System, Object Oriented Programming, Structured Programming Language